Fat Containers and Multipass: Building a Cloud on Your Laptop

Learn how fat containers and the Multipass tool can help create cloud virtual machine stacks on your laptop, thus accelerating productivity while cutting cloud costs.

The concept of partitioning static hardware resources into dynamic units of computing through virtual machines (VMs) revolutionised how businesses ran IT. Similarly, further dynamic repartitioning of user space through containers is revolutionising how businesses are running modern distributed infrastructure.

The principles of Dev(Sec)Ops are based on the good old test driven development (TDD) methodologies, which state that we need to consider how we'll test first before even writing a single line of code. Today, it's all about 'fail early and fail often, and continuous improvements'.

Therefore, the ability to bring up an environment consisting of different VMs running various software is today a necessity for on-demand development and testing (also as a vital part of CI/CD). Waiting to get VMs in data centers (DCs) takes at least a few days and public cloud-based dev/QA/POC accounts are subject to various limitations and approvals, especially if a company is spending substantial money on public cloud infrastructure.

Fat container VMs

Virtual machines were always about emulating the whole machine through a hypervisor taking control of the entire underlying hardware. This makes VMs heavyweight entities requiring more computing resources. Containers are lightweight as the devices and kernel are shared and only the user space is sandboxed. While they require VMs to run on (they can run directly on hardware too) they can be nested better than VMs, at least for dev/QA/POC purposes.

If you have used a combination of Vagrant and VirtualBox, you may have realised that the virtualisation provided is heavyweight, requiring special hardware acceleration and significant computing resources. In my experience, only four to five VMs are enough to make a 4 core/16GB